THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Assertion (A): Rusting of iron is a chemical change. Reason (R): A new substance with different properties (iron oxide) is formed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Dry HCl gas does not change the colour of dry blue litmus paper. Reason (R): HCl gas does not ionise in the absence of water to release H^+ ions. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Sodium is stored in kerosene oil. Reason (R): Sodium reacts vigorously with oxygen and moisture of the air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): The rate of breathing in aquatic organisms is much faster than in terrestrial organisms. Reason (R): The amount of dissolved oxygen in water is fairly low compared to air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): Reflex actions are not controlled by the brain directly. Reason (R): The reflex arc connects the receptor to the effector through the spinal cord. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Asexual reproduction produces offspring that are not exactly identical to the parent. Reason (R): Copying of DNA is accompanied by small errors. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): Sex of a child is determined by the father. Reason (R): Sperms carry either X or Y chromosome. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A convex mirror is used as a rear-view mirror. Reason (R): It always forms virtual, erect and diminished images with a wide field of view. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The image formed on the retina is inverted. Reason (R): The brain interprets the signals to make the object appear erect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): A voltmeter is connected in parallel with a resistor. Reason (R): A voltmeter has a very high resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): Magnetic field lines never intersect. Reason (R): The field at a point has only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): The number of trophic levels in a food chain is limited. Reason (R): About 90 per cent of energy is lost at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 28 g **(b)** 20 g **(c)** 12 g **(d)** 44 g

15. Equal volumes of HCl and NaOH solutions of the same concentration are mixed. The resulting solution has pH: 1

(a) 13 **(b)** 1 **(c)** 3 **(d)** 7

16. Metal X reacts with dilute HCl giving H_2 but does not react with cold or hot water. Metal Y reacts with cold water. Which is more reactive? 1

(a) Y **(b)** cannot be said **(c)** both equally **(d)** X

17. **Assertion (A):** Burning of magnesium ribbon in air is a combination reaction. 1
Reason (R): In a combination reaction two or more substances combine to form a single product.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Dry hydrogen chloride gas does not change the colour of dry blue litmus paper. 1
Reason (R): Acids produce H^+ ions only in the presence of water.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** Sodium is stored under kerosene oil. 1
Reason (R): Sodium reacts vigorously with oxygen and moisture of the air and may catch fire.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Carbon forms a very large number of compounds. 1
Reason (R): Carbon is tetravalent and shows the property of catenation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. 12 g of carbon burns completely in 32 g of oxygen. Find the mass of carbon dioxide formed and name the law applied. 2

22. When a few drops of conc. HCl are put on a piece of limestone, bubbles form. Name the gas, test and equation. 2

23. A metal X reacts with dilute HCl and also with NaOH producing hydrogen. Identify the nature of X's oxide with an example metal. 2

24. Write the balanced equation for the complete combustion of ethane and name the products. 2

25. Calculate the volume of blood pumped in 1 minute if the heart beats 72 times per minute and pumps 70 mL each time. 2

26. A patient cannot pull his hand away from a hot object but can feel the heat. Which part of the reflex arc is damaged? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Balance $\mathrm{PbO} + \mathrm{C} \longrightarrow \mathrm{Pb} + \mathrm{CO}$ and identify the oxidising and reducing agents. 3

28. A student is given solutions of pH 1, 4, 7, 10 and 13. Which will turn phenolphthalein pink and which can neutralise a base? Arrange by acidity. 3

29. In the reaction $\mathrm{ZnO} + \mathrm{C} \longrightarrow \mathrm{Zn} + \mathrm{CO}$ identify the oxidising agent, reducing agent and the type of reaction. Why is it used for zinc? 3

30. Write the structural formula of propene and state two ways in which it differs from propane. 3

31. In an experiment, the destarched leaf was covered with black paper partly. After sunlight and iodine test the uncovered part turned blue-black. Interpret fully. 3

OR

A seedling is kept in a box with a window on one side; after a few days it bends. When the tip was cut, no bending was observed. Explain. 3

32. After 2 hours, the number of Amoebae from one cell that divides every 15 minutes is calculated. Find it and state the type of reproduction. 3

OR

A pea plant has yellow round seeds. How would you find its genotype? Describe the cross and the possible results. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** In a school laboratory, Rohan took about 2 g of green ferrous sulphate crystals in a dry boiling tube and heated it strongly. The crystals first lost water and turned white, then changed to a reddish-brown solid, and a smell like burning sulphur was noticed. 5

(i) What is the colour of the final solid residue?

(ii) Name the type of reaction taking place on strong heating.

(iii) Write the balanced chemical equation for the decomposition of anhydrous ferrous sulphate.

(iv) Name the two gaseous products and state whether the reaction is endothermic or exothermic, with a reason.

34. **Case study.** A gardener, Harpal, tested four soil-water samples with a pH paper. Sample P showed pH 5, sample Q pH 7, sample R pH 9 and sample S pH 13. 5

(i) Which sample is neutral?

(ii) Which sample is the strongest base?

(iii) Which sample(s) will turn red litmus blue?

(iv) Harpal wants to grow a crop in the acidic sample P. Name a substance he can add to the soil to bring the pH nearer to 7 and give the reason.

35. Four metals A, B, C and D: A displaces B, C and D from their salt solutions; B displaces C and D; C displaces D; D displaces none. Arrange them in order of reactivity and suggest a suitable extraction method for A and for D. Name one possible set of four metals. 5

36. Explain conversion of ethanol to ethene, ethanoic acid and ethyl ethanoate with reagents and equations. Identify the oxidising agent and the catalyst. 5

OR

A student sets up an experiment: Leaf in a flask with KOH, another with water, both in sunlight; iodine test is done. State the aim, result and conclusion; also give a control. 5

37. Design an experiment to show that tip of the shoot is necessary for phototropism. State aim, procedure, observation and conclusion. 5

OR

Explain, with reference to variations, why sexual reproduction is more advantageous to a species than asexual reproduction, giving an example from plants. 5

38. A pea plant homozygous for tall and round seeds is crossed with a dwarf plant having wrinkled seeds. Show the F1 and the F2 with the number of each phenotype in 1600 F2 plants, and the genotype of the dwarf round plants. 5

OR

An object 4 cm tall is placed at 30 cm from a concave mirror of f = 20 cm. A convex mirror of f = 20 cm is then used in the same place. Compare the images in the two cases (position, size, nature). 5

END OF QUESTION PAPER — Total: 80 Marks